RULE #0: READ THIS BEFORE YOU TOUCH PRUSASLICER (ZERO-ERROR GUARANTEE)

1. DO NOT ROTATE OR SCALE ANY FILE! Every single .stl file in FINAL_PRINT_FILES/ is ALREADY pre-rotated and sitting 100% flat on the print bed (Z = 0.00 mm). Just drag the file into PrusaSlicer and leave its position/rotation alone!
2. SUPPORTS = 'NONE' FOR EVERY SINGLE FILE! Do NOT turn on supports. Every overhang and screw hole is self-supporting.
3. ALL SCREW HOLES ARE ALREADY 3D-PRINTED! You do NOT need a drill or tap. Even better: File #6 prints 12 click-in Snap-Lock Pins so you can assemble the entire clock by hand with ZERO metal screws and ZERO tools!
4. EXTRA-DEEP 41.0mm BACK CABLE VAULT: The frame & motor plate have deep recessed pockets + 6 printed wire-tuck hooks specifically designed to hold all your loose individual Dupont jumper wires & breadboard without touching the back cover.

THE 6 FILES IN 'FINAL_PRINT_FILES/' — WHICH PRINTER & WHICH FILAMENT SPOOL

FILE NAME (IN FINAL_PRINT_FILES/)	PRINTER	FILAMENT SPOOL	SLICER PRESET	TIME
1_Outer_Frame_MATTE_WHITE_PrusaMK4.stl □ Deep 52mm frame (41mm rear cable room!). Just slice & print.	Prusa MK4 #1	Overture Matte White	0.20mm STRUCTURAL □ 15% Gyroid	2h 48m
2_Dial_Face_WHITE_then_BLACK_at_Z2.6mm_PrusaMK4.stl □ Starts in Matte White; printer pauses at Z=2.60mm for Matte Black!	Prusa MK4 #2	White -> Black @ Z=2.6mm	0.20mm STRUCTURAL □ Color Change	1h 16m
3_Motor_Mount_Plate_ORANGE_or_WHITE_PrusaMK4.stl □ Has deep Breadboard/ULN2003 pockets + 6 Dupont cable clips!	Prusa MK4 #2	Fortis Orange #58 (or White)	0.20mm STRUCTURAL □ 20% Infill	58m
4_Back_Cover_TRANSPARENT_or_WHITE_PrusaMK4.stl □ Exhibition clear backplate with keyhole wall slot & USB-C notch.	Prusa MK4 #2	Fortis Transparent #126	0.20mm STRUCTURAL □ 100% Infill	44m
5_Pointer_Hands_MATTE_ORANGE_PrusaMINI.stl □ Prints both hands (40.25mm Beach Hand + 25.30mm WORTH IT? hand).	Prusa MINI	Fortis Matte Orange #58	0.15mm STRUCTURAL □ 100% Infill	14m
6_Snap_Lock_Pins_No_Screws_Needed.stl □ 12x barbed click-in pins (8 long frame pins + 4 short motor pins).	Prusa MINI	Matte Orange or White	0.15mm STRUCTURAL □ 100% Infill	9m

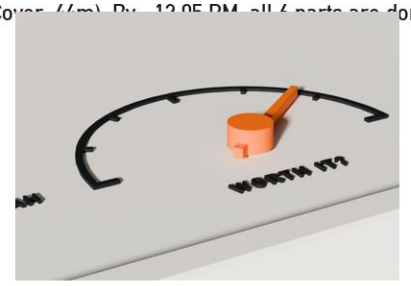
PARALLEL FLEET TIMELINE — FINISH EVERYTHING IN UNDER 3 HOURS!

- AT 9:00 AM — START ALL 3 PRINTERS AT THE SAME TIME:
  1) On Prusa MK4 #1: Load Overture Matte White PLA. Start File #1 (1_Outer_Frame). It will run for 2h 48m untouched.
  2) On Prusa MK4 #2: Load Overture Matte White PLA. Start File #2 (2_Dial_Face). At ~10:12 AM (Z = 2.60 mm), MK4 #2 will beep and ask you to swap in Overture Matte Black PLA for the final 6 minutes of raised lettering!
  3) On Prusa MINI: Load Fortis Matte Orange #58 PLA. Print File #5 (Hands, 14m) and File #6 (Snap-Lock Pins, 9m).

- AT 10:20 AM — WHEN PRUSA MK4 #2 FINISHES THE DIAL FACE:
  4) Remove the finished Dial Face from MK4 #2. Load Fortis Matte Orange #58 PLA (so the internal motor plate looks awesome through the transparent back!) and print File #3 (3_Motor_Mount_Plate, 58m).

- AT 11:20 AM — FINAL PRINT ON PRUSA MK4 #2:
  5) Load Fortis Pro Transparent #126 PLA on MK4 #2 and print File #4 (4_Back_Cover, 44m). By 12:05 PM, all 6 parts are done!

□ MACRO CAD PREVIEW OF FILE #2 & #5:
0.60mm Raised Matte Black DIN Text,
100% Connected L-Corners on Gauge,
and Capped Matte Orange Needle Hub.



A. HOW TO SLICE REGULAR FILES (#1, #3, #4, #5, #6) — 4 SIMPLE CLICKS

STEP 1: Open PrusaSlicer on your Mac. Click 'File' -> 'Import' -> 'Import STL/3MF...' (or just drag the .stl file onto the platter).

STEP 2: Look at the right-hand sidebar in PrusaSlicer and set the 4 dropdown menus at the top:

- Print settings: Choose '0.20mm STRUCTURAL' (on Prusa MK4) OR '0.15mm STRUCTURAL' (on Prusa MINI for Hands/Pins).
- Filament: Choose 'Generic PLA' (or 'Prusament PLA' — both use 215°C nozzle / 60°C bed, perfect for Overture & Fortis).
- Printer: Choose 'Original Prusa MK4 Input Shaper 0.4 nozzle' (or 'Original Prusa MINI+ Input Shaper 0.4 nozzle').
- Supports: Choose 'None' <-- CRITICAL! Never enable supports on any of these 6 files!
- Infill: 15% or 20% for Frame #1 and Plate #3. Set Infill to 100% for Back Cover #4, Hands #5, and Snap Pins #6.

STEP 3: DO NOT rotate or move the object! It is already lying flat on Z = 0.00 mm.

STEP 4: Click 'Slice now' (bottom right), then click 'Export G-code' to save to your USB stick (or send via Prusa Connect).

B. IDIOT-PROOF GUIDE: HOW TO DO THE WHITE-TO-BLACK SWAP ON FILE #2 (DIAL FACE)

Why we do this: File #2 (2_Dial_Face_WHITE_then_BLACK_at_Z2.6mm_PrusaMK4.stl) is a single solid piece where the flat dial face is 2.40 mm thick (Layers 1 to 12) and the letters & gauge ticks stick up from 2.40 mm to 3.00 mm (Layers 13, 14, and 15).

By clicking ONE button in PrusaSlicer, the printer will print the flat face in WHITE, then pause and beep so you load BLACK!

CLICK 1: Drag '2_Dial_Face_WHITE_then_BLACK_at_Z2.6mm_PrusaMK4.stl' into PrusaSlicer (0.20mm STRUCTURAL, Supports: None).

CLICK 2: Click the big 'Slice now' button in the bottom-right corner.

CLICK 3: Look at the vertical ORANGE SLIDER BAR on the right side of the 3D preview window (it shows layer height in mm).

Click and drag the TOP thumb of that vertical orange bar DOWN until the height box reads '2.60' (Layer 13).

(You will see the first slice of the letters LONG REEF, DEE WHY, and WORTH IT? appear on screen at 2.60 mm!)

CLICK 4: Click the little ORANGE '+' (PLUS SIGN) icon right next to the '2.60' box on the slider bar.

PrusaSlicer immediately inserts a Color Change (M600) pause at 2.60 mm and turns the top 3 layers a different color in preview!

CLICK 5: Click 'Slice now' one final time, then click 'Export G-code'!

WHAT TO DO AT THE PRINTER:

- Load Overture Matte White PLA into Prusa MK4 #2 and start the print. Walk away for 1 hour 12 minutes!
- At Z = 2.60 mm, the MK4 will beep, park the nozzle at the front, and push out the White filament. Pull the White spool out, push Overture Matte Black PLA into the extruder hole, confirm on screen that the color is clean black, and click 'Continue'!

C. PRO TIPS SO YOUR PRINT LOOKS LIKE A $600 BRAUN GALLERY PIECE (NOT 3D PRINTED)

1. CLEAN THE STEEL SHEET WITH RUBBING ALCOHOL (IPA): Fingerprints on the Prusa bed cause corners to lift. Wipe the Satin or Smooth PEI sheet with a paper towel and Isopropyl Alcohol right before starting File #1 and File #2.
2. TURN ON 'IRONING' (OPTIONAL BONUS FOR FILE #2 DIAL FACE): If you want the white dial surface and black letter tops to feel like injection-molded ceramic, go to 'Print Settings' -> 'Infill' -> 'Ironing' -> check 'Enable ironing' -> 'All top surfaces'! (Adds ~18 minutes to File #2 and melts the top surface with a hot pass so layer lines disappear completely!)
3. TRANSPARENT BACK COVER (#4) GLASS EFFECT: When slicing File #4 in Fortis Pro Transparent #126 PLA, set 'Infill' -> 'Fill density' to '100%' and 'Fill pattern' to 'Aligned Rectilinear'. All internal lines align like ribbed architectural glass!

WHERE DO THE SCREWS OR SNAP-PINS GO? (100% PRE-PRINTED HOLES!)

- YOU DO NOT NEED TO DRILL OR MAKE ANY HOLES! Every single hole is ALREADY 3D-printed directly into the parts:
  - The Outer Frame (#1) has 4 corner pillars with Ø2.85 mm pre-printed holes on the inside.
  - The Motor Mount Plate (#3) has 4 corner holes (Ø3.20 mm) + 4 motor ear holes (Ø2.85 mm) pre-printed.
  - The Back Cover (#4) has 4 corner holes (Ø3.40 mm) pre-printed.
- TWO WAYS TO FASTEN IT TOGETHER (CHOOSE WHICHEVER YOU PREFER):

OPTION A — 100% TOOL-FREE (USE FILE #6 '6_Snap_Lock_Pins_No_Screws_Needed.stl'):

File #6 prints 12 barbed plastic snap-pins (8 Longer Pins for the 4 corners of Plate #3 and Cover #4, plus 4 Shorter Pins for the motor ears). You literally just push the pin through the hole with your thumb until it clicks snug!

OPTION B — STANDARD M3 SCREWS (IF YOU HAVE AN M3 SCREW ASSORTMENT BOX):

Thread 4x M3×8mm screws through Plate #3, 4x M3×6mm screws into the motor ears, and 4x M3×8mm screws through Back Cover #4.

THE SANDWICH ASSEMBLY ORDER (LAY IT FACE-DOWN ON A SOFT TOWEL!)

1

LAYER 1 (BOTTOM ON TOWEL): 1_Outer_Frame

Lay FACE-DOWN on a soft towel so the deep 52mm cavity faces UP at you.

2

LAYER 2: 2_Dial_Face (White + Black Text)

Drop FACE-DOWN into the frame! The black letters face DOWN toward the table.

3

LAYER 3: 3_Motor_Mount_Plate (Orange)

Drop onto the back of the Dial Face (recessed pockets & cable clips face UP at you!).
Push 4 Long Snap-Pins (or 4x M3x8mm screws) into the 4 corners.

4

LAYER 4: 2x 28BYJ-48 Stepper Motors

Drop the 2 silver motors into the circular seats on Plate #3 so their gold shafts poke down through the 2 holes in the dial face! Secure ears with 4 Short Snap-Pins.

5

LAYER 5: ESP32-S3 + ULN2003 + Dupont Wires

Drop Breadboard/ESP32 into the LEFT deep pocket & 2x ULN2003 boards into the RIGHT deep pocket. Tuck all loose Dupont jumper wires under the 6 printed clips!

6

LAYER 6 (TOP): 4_Back_Cover (Transparent)

Route USB-C cable out the bottom slot. Place Transparent Back Cover on top and click in the final 4 Long Snap-Pins (or 4x M3x8mm screws)!

HOW WE ENGINEERED 41.0 MM (1.61 INCHES!) OF VERTICAL ROOM FOR YOUR CABLES

You mentioned you are using individual female/male Dupont jumper wires (not flat ribbon PCBs) and need plenty of room in the back so nothing gets squished under the Transparent Back Cover (#4). Here is what is built into your STL files:

1. EXTRA-DEEP 52.0 MM OUTER FRAME (#1): Gives you 41.0 mm (1.61 inches) of internal height behind the recessed electronics bays! Even a 10mm Mini Breadboard + 11mm ESP32-S3 + 14mm upright plastic Dupont connector pins (35mm total) has 6.0 mm of empty air above the pins before the Transparent Back Cover (#4) even touches down!

2. TWO DEEP RECESSED ELECTRONICS CRADLES IN PLATE #3 (3_Motor_Mount_Plate):
   - LEFT DEEP POCKET (84 × 44 mm, recessed down to Z = 1.8mm): Fits your Mini Breadboard + ESP32-S3 deep inside the plate. Stick a small square of double-sided foam tape (or Blu-Tack) under the breadboard and press it into the Left Pocket.
   - RIGHT DEEP POCKET (84 × 44 mm, recessed down to Z = 1.8mm): Fits BOTH of your ULN2003 Stepper Driver Boards side-by-side! Stick both ULN2003 boards into the Right Pocket using double-sided foam tape.

3. 6 BUILT-IN 'SPAGHETTI-TAMER' L-HOOK WIRE CLIPS PRINTED AROUND THE POCKETS:
   - Once you connect your 12 individual Dupont wires between the ESP32-S3 and the two ULN2003 boards, simply gather the loose middle loops of wire and push them UNDER the 6 printed L-shaped hook clips around the edges of the pockets!
   - Those 6 hooks hold every single wire flat at Z = 34mm (15mm below the transparent back cover!). Zero wires pop up!

PIN-BY-PIN DUPONT WIRING CHEAT SHEET (ESP32-S3 TO 2x ULN2003 BOARDS)

- UPPER MOTOR (BEACH HAND — PLUGS INTO ULN2003 BOARD #1 WHITE SOCKET):
  - ULN2003 #1 'IN1' ---> Dupont Wire to ESP32-S3 GPIO 4
  - ULN2003 #1 'IN2' ---> Dupont Wire to ESP32-S3 GPIO 5
  - ULN2003 #1 'IN3' ---> Dupont Wire to ESP32-S3 GPIO 6
  - ULN2003 #1 'IN4' ---> Dupont Wire to ESP32-S3 GPIO 7

- LOWER MOTOR ('WORTH IT?' GAUGE — PLUGS INTO ULN2003 BOARD #2 WHITE SOCKET):
  - ULN2003 #2 'IN1' ---> Dupont Wire to ESP32-S3 GPIO 15
  - ULN2003 #2 'IN2' ---> Dupont Wire to ESP32-S3 GPIO 16
  - ULN2003 #2 'IN3' ---> Dupont Wire to ESP32-S3 GPIO 17
  - ULN2003 #2 'IN4' ---> Dupont Wire to ESP32-S3 GPIO 18

- POWER WIRES (4 DUPONT WIRES VIA MINI BREADBOARD + / - RAILS):
  - ESP32-S3 '5V' (or VBUS) ---> Breadboard Red (+) Rail ---> Both ULN2003 '+' Pins
  - ESP32-S3 'GND' ---> Breadboard Blue (-) Rail ---> Both ULN2003 '-' Pins

FINAL STEP: PRESSING ON THE 2 ORANGE HANDS (10 SECONDS, NO GLUE NEEDED!)

1. Flip your assembled clock FACE-UP on the table (BEFORE plugging in the USB-C power cord).
2. Look at the two gold motor shafts sticking out 5.40 mm in front of the white dial face. Notice each gold shaft has two flat sides ('D-shaped double-flat shaft'), and the inside of the 3D-printed Orange Hands has the exact matching D-shaped hole + a solid closed cap on top (so the metal shaft is completely hidden!).
3. Take the LONG ORANGE HAND (40.25 mm) -> align it pointing straight UP at 'LONG REEF' (12 o'clock) and press it firmly down onto the UPPER motor shaft with your thumb until it stops (leaves a clean ~1.2 mm floating shadow gap above the dial).
4. Take the SHORT ORANGE HAND (25.30 mm) -> align it pointing straight UP at the center 12 o'clock tick of the 'WORTH IT?' semi-circle and press it firmly down onto the LOWER motor shaft. Plug in USB-C — your Surf Clock is alive!

HOW TO PRINT THE 4 STUDIO COLORWAY VARIATIONS WITH YOUR EXISTING PLA SPOOLS



CLASSIC RAMS + EXHIBITION CLEAR BACK

Frame #1: Matte White □ Dial #2: White->Black
Plate #3: Orange #58 □ Cover #4: Transparent #126



VARIANT A: 'REVERSE BRAUN / PANDA'

Frame #1: Matte White □ Dial #2: Start BLACK, swap to WHITE at Z=2.60mm! □ Hands #5: Orange #58



VARIANT B: 'SKELETON TRANSPARENT DIAL'

Frame #1: Matte White □ Dial #2: Start Transparent #126, swap to BLACK at Z=2.60mm! □ Plate #3: Orange



VARIANT C: 'STEALTH MATTE BLACK FRAME'

Frame #1: Overture Matte Black □ Dial #2: White then Black at Z=2.60mm □ Hands #5: Orange #58



VARIANT D: 'SIGNAL ORANGE OB-4 EDITION'

Frame #1: Fortis Matte Orange #58 □ Dial #2: White then Black at Z=2.60mm □ Hands #5: Orange #58



FLAT-LAY STUDIO PROPORTIONS (184 × 230 × 52 mm)

Fits Prusa MK4 with 20mm margin to spare and houses all individual Dupont jumper cables cleanly.